The basic properties of a complete Sigma-consistent set of formulas

The Open Logic Project contributors

openlogic:normal-modal-logic/completeness:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author's own proposition environment prop:ccs-properties (source lines 504-528); the proof environment begins two lines below it and is its own proof, closing at source line 597.

Editorial note. The statement's individual clauses carry their own internal labels (prop:ccs-closed and the tag items); they lie inside the counted statement span, so the proof's references to them resolve inside the package and neither a context span nor a source pointer is needed. No other reference and no citation occurs in the retained span.

Editorial note. Several proof clauses are written with the author's \iftag switch, whose first branch is a placeholder for an exercise version of the text; with the shipped preamble the second branch is the one typeset, so the package carries the author's full argument as intended for this unit.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit (it declares the theorem-like environments, the shorthand accents and the tag-literal handling macros). Author text is untouched; only the packaging is the intake line's.

Editorial note. Section and theorem counters are reset to zero so the retained proposition prints with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's normal modal logic completeness unit, counted once: the property list of a complete Sigma-consistent set, including the mirroring of every propositional connective.

Allowed standard prerequisites (editorial).

- Complete Sigma-consistent set. A set Gamma of formulas is Sigma-consistent when it does not prove falsity in the calculus with axioms Sigma, and complete when for every formula A either A or its negation belongs to Gamma.. Both defining properties are used in every clause, most visibly in the negation case and in the consistency arguments (source lines 505-506, 533-536, 550-552).
- Deductive closure. Gamma is deductively closed in Sigma, so any formula provable from Gamma together with Sigma belongs to Gamma.. The first clause proves deductive closure and the remaining clauses use it repeatedly

to move from a proof to membership (source lines 532-537, 538-541, 546-548, 554-561).

- Tautological instances and their provability. Every tautological instance of a propositional schema is provable from Sigma, and combining such an instance with membership in Gamma yields further membership.. Each connective clause invokes a specific tautological instance, for example the introduction and elimination schema for conjunction or the extraction schema for implication (source lines 554-561, 563-568, 570-579, 581-595).

Proposition 0.1. *Suppose* Γ *is complete* Σ*-consistent. Then:*

1. Γ *is deductively closed in* Σ*.*

2. $\Sigma \subseteq \Gamma$*.*

prvFalse $\bot \notin \Gamma$

prvTrue $\top \in \Gamma$

3. $\neg !A \in \Gamma$ *if and only if* $!A \notin \Gamma$*.*

prvAnd $!A \wedge !B \in \Gamma$ *iff* $!A \in \Gamma$ *and* $!B \in \Gamma$

prvOr $!A \vee !B \in \Gamma$ *iff* $!A \in \Gamma$ *or* $!B \in \Gamma$

prvIf $!A \rightarrow !B \in \Gamma$ *iff* $!A \notin \Gamma$ *or* $!B \in \Gamma$

prvIff $!A \leftrightarrow !B \in \Gamma$ *iff either* $!A \in \Gamma$ *and* $!B \in \Gamma$*, or* $!A \notin \Gamma$ *and* $!B \notin \Gamma$

Proof. 1. Suppose $\Gamma \vdash [\Sigma]!A$ but $!A \notin \Gamma$. Then since Γ is complete Σ-consistent, $\neg !A \in \Gamma$. This would make Γ inconsistent, since $!A, \neg !A \vdash [\Sigma]\bot$.

2. If $!A \in \Sigma$ then $\Gamma \vdash [\Sigma]!A$, and $!A \in \Gamma$ by deductive closure, i.e., case 1.

prvFalse If $\bot \in \Gamma$, then $\Gamma \vdash [\Sigma]\bot$, so Γ would be Σ-inconsistent.

prvTrue $\Gamma \vdash [\Sigma]\top$, so $\top \in \Gamma$ by deductive closure, i.e., case 1.

3. If $\neg !A \in \Gamma$, then by consistency $!A \notin \Gamma$; and if $!A \notin \Gamma$ then $!A \in \Gamma$ since Γ is complete Σ-consistent.

prvAnd Exercise.

prvOr Exercise.

prvIf Exercise.

prvIff Exercise.

□